SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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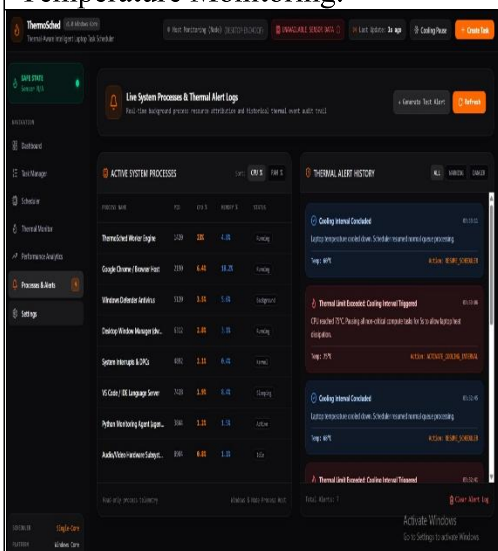
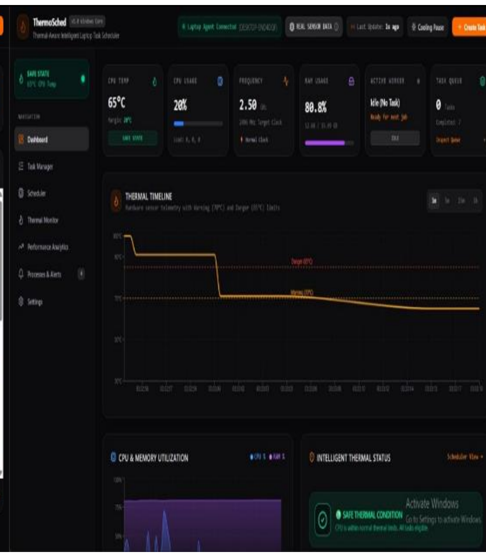
Course Code: CSA1215 **Slot:** A

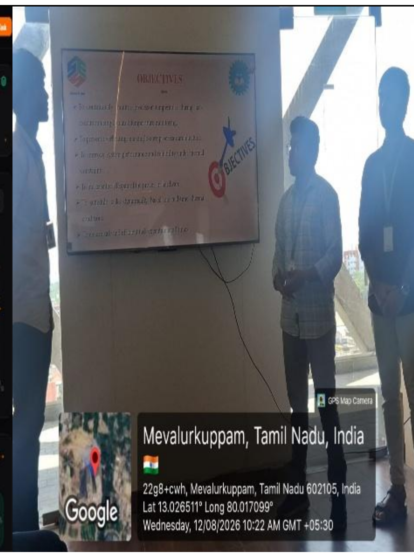
Course Name: Computer Architecture

Course Faculty: Dr. Santhakumar D

Project Title: Thermal-Aware Task Scheduling for Multi-Core Processor Architecture Using Dynamic Temperature Monitoring.







Project Description: (here you write what you did in this project (contribution) including Model Description

This module focuses on **continuously monitoring the performance and temperature of each CPU core** in the multi-core processor.

- Collects CPU **core temperature, utilization, frequency, and workload** information.
- Monitors the thermal condition of individual cores in real time.
- Identifies cores that are **cool, moderately loaded, or thermally stressed**.
- Stores the collected processor data for further analysis.
- Provides the thermal and performance information required by the scheduling module.
- Helps prevent excessive temperature buildup and potential thermal throttling.

Output: Real-time processor performance and temperature data.

Module 1: Processor Performance and Thermal Monitoring —Study multi-core processor architecture, monitor CPU utilization, core temperature, frequency, workload and thermal status. Develop the thermal/performance data collection component. Verify sensor/data collection accuracy; analyze processor utilization and temperature variations under different workloads.

Student Signature **Guide Signature**